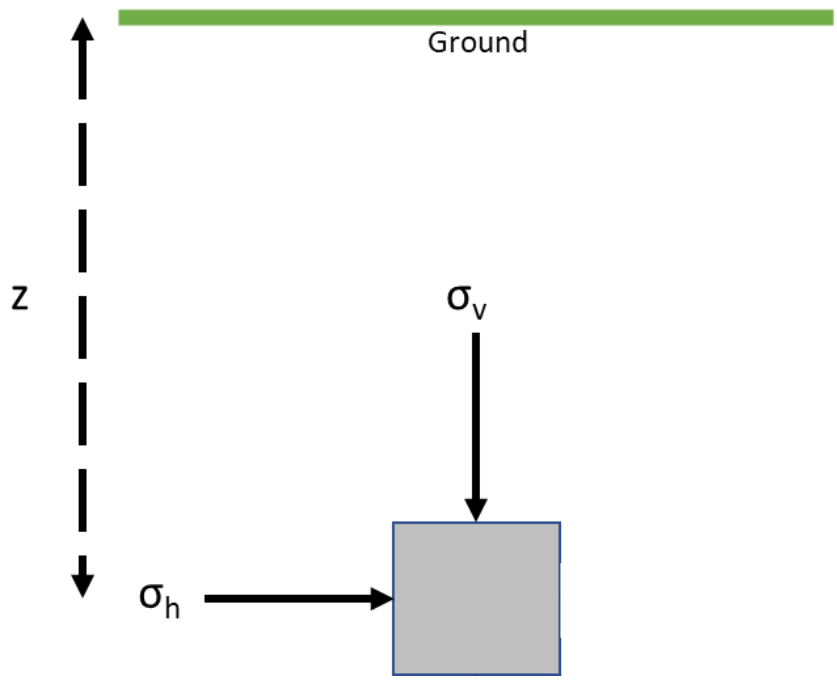
MNTC 313 – Assignment 7

In underground mining, it is important to account for stress fields in the surrounding rock when developing an excavation. The type and magnitude of stress surrounding an underground excavation will necessitate varying degrees of artificial ground support to prevent a collapse. Stress models are often complex, as stresses are irregular and fluctuate based on the rock mass and proximity to other excavations. For this assignment, we will only analyze the vertical and horizontal stresses acting at certain depths below the ground.



Vertical stress is dependent on the weight of rock above the point of analysis. Vertical stress can be found using the following formula:

$$\sigma_v = \gamma z$$

Where γ is the unit weight of the rock in kN/m^3, and z is the depth at a given point in metres. The resulting value is in kilopascals, kPa.

Horizontal stresses commonly fluctuate relative to vertical stresses. For simplicity, we will assume that the horizontal stresses are always related to the vertical stresses by a constant, k.

$$\sigma_h = k\sigma_v,$$

however, it is difficult to measure k with certainty so different values of k should be tried.

MNTC 313 – Assignment 7

Write a program in C/C++ that finds the horizontal and vertical stresses at equally spaced depths below the ground:

- Ask for user input of the rock unit weight, γ, the number of depth values, *Nd*, the first depth value, and the increment between the depth values, the number of *k* values, *Nk*, the first value for the constant, *k*, and the increment between the *k* values.
- Using the inputted data, build a row array of the depth values, a row array of the vertical stresses, a row array of the *k* values and an array of the horizontal stresses. Print the vertical stress values in a single column.
- Write a file named "outputFileA.txt" that contains the depth values in the first column and horizontal stress values in the adjacent columns, separated by commas.

Note: Your file output format should match the expected file output included. **After submitting to Gradescope, submit an example of your output file to the OnQ dropbox along with a .cpp of your program**. TAs will be checking programs to make sure that the appropriate code to generate the output files is present.

Reminder: Please **DO NOT** include any personal information such as name, student number or Net ID in your submission. All submissions to Gradescope will be run through plagiarism detection software. The assignment is due at 11:59 PM ET on Sunday of Week 7. **Late submissions are not accepted**.

Your output must look like:

```
Input rock unit weight (kN/m3):
10
Input number of depth values:
10
Input first depth value (m):
10
Input depth increment (m):
1
Input number of k values:
10
Input first k value (m):
1
Input k increment (m):
1
Vertical stress values (kPa):
100
110
120
130
140
150
160
170
180
190
```

MNTC 313 – Assignment 7

Your file must look like:

```
10,100,200,300,400,500,600,700,800,900,1000
11,110,220,330,440,550,660,770,880,990,1100
12,120,240,360,480,600,720,840,960,1080,1200
13,130,260,390,520,650,780,910,1040,1170,1300
14,140,280,420,560,700,840,980,1120,1260,1400
15,150,300,450,600,750,900,1050,1200,1350,1500
16,160,320,480,640,800,960,1120,1280,1440,1600
17,170,340,510,680,850,1020,1190,1360,1530,1700
18,180,360,540,720,900,1080,1260,1440,1620,1800
19,190,380,570,760,950,1140,1330,1520,1710,1900
```

Depth Horizontal stress for different values of *k* given depth in first column